FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2016

ANSWERS, HINTS & SOLUTIONS

FULL TEST – I
(PAPER-2)

No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	C	B
2.	A	B	A
3.	B	C	C
4.	C	A	A
5.	C	B	B
6.	C	A	C
7.	D	A	A
8.	A, B	A, B	A, B
9.	B, C	B, D	A, C
10.	C, D	B, C	B, D
11.	A, D	A, B, C	A, B
12.	D	D	B
13.	C	C	C
14.	B	D	A
15.	C	A	B
16.	D	C	C
1.	(A) → p,q, (B) → p,q, (C) → p,q,s, (D) → r,s	(A) → p,r,s, (B)→p,r,s,t, (C) → q, r, (D) → p, r, s	(A) → p, q, r, s (B) → p, r, s (C) → t (D) → p, s
2.	(A) → r, (B) → t, (C) → q, (D) → r	(A)→p,s, (B) → q,r,t, (C) → p,s, (D) →q,r,t	(A) → q, (B) → p, (C) → r, s, (D) → r
1.	4	3	8
2.	8	2	6
3.	1	6	4
4.	3	2	7
5.	1	6	3
6.	0	4	2

Physics

PART – I

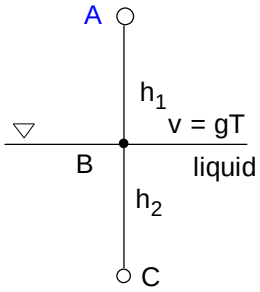
SECTION – A

1. $$Q = \frac{\Delta\phi}{R} = \frac{\text{charge in flux}}{\text{resistance}}$$
$$= \frac{(B \times A \times n)}{R} = \frac{\mu_0}{4\pi}\left(\frac{2\pi I}{r}\right) \times A \times n \times \frac{1}{R}$$
$$= \left(\frac{\mu_0 IAn}{2rR}\right)$$

2. Relative acceleration of rod with respect to wedge is parallel to the inclined plane.

3. $$f_{app} = \left(\frac{v - v_D}{v - f_s}\right) f_0$$
Simply use the above relation

4. for B to C part



$u = gT$

$$a = -\frac{(v \times \rho - v \times \rho_0)g}{v\rho_0}$$

$$v = 0 \quad a = \frac{T}{2}$$

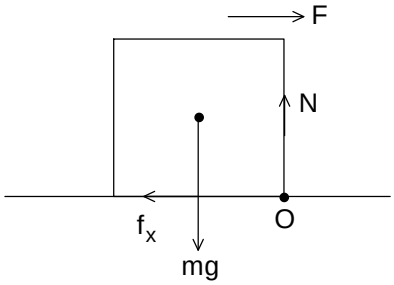
$$O = gt - \left(\frac{\rho - \rho_0}{\rho_0}\right) g \times \frac{T}{2}$$

$$1 = \frac{\rho - \rho_0}{2\rho_0} \Rightarrow 2\rho_0 = \rho - \rho_0$$

$$\frac{\rho}{\rho_0} = 3$$

5. Calculate the initial and final distance of object from the concave mirror.

6. To avoid t toppling



$(\tau_{net})_0 = 0$

$F \times a - mg \times \frac{a}{2} = 0$

$F = \frac{mg}{2}$

Now apply Newton's laws of motion.

7. As we know
$L_0 = L_1 + L_2$
L_1 = due to rotational motion
L_2 = due to linear motion.

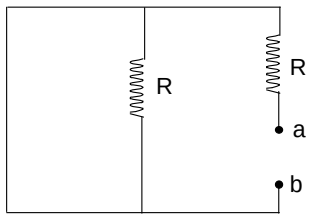
8. If temperature of a body is constant that means amount of hear energy received and amount of hear energy emitted are equal.

9. If direction of field is along the x-axis then its magnitude will be 20 V/cm other wise greatest then hat.

10. Apply conservation of linear momentum

$KE = \frac{\rho^2}{2 \times mass}$

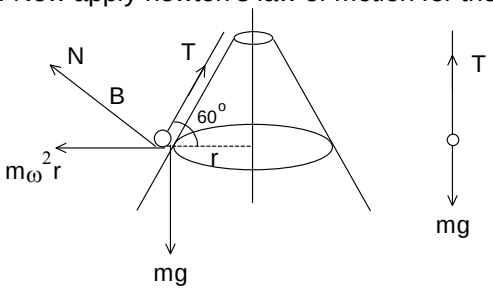
11. Calculate the time constant



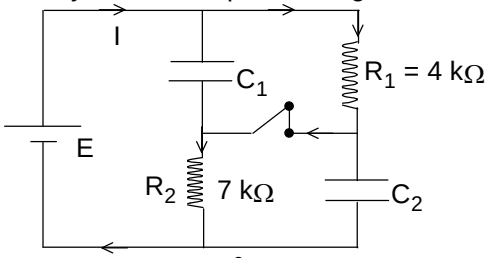
Battery is replaced by a wire

$\tau = \frac{L}{R_{ab}} = \frac{L}{R}$

12 – 13. Now apply newton's law of motion for the particle B.



14 – 16. Initially current will pass through the resistance only.



Power in $R_2 = 7 \times (I^2) = 2.8$ W

Power in $R_1 = 4 \times I^2 = 4 \times (4) = 1.6$

Charge in $C_1 = 3 \times 10^{-6} \times 8 \times 10^3 \times \sqrt{\frac{2.8}{7 \times 10^3}}$

$= 240\ \mu C.$

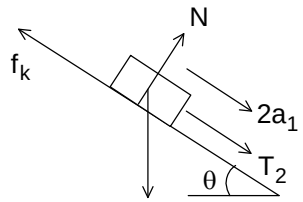
SECTION – B

1. In such type of situation, we have to use these basic facts.
(a) charge will flow between the two body if there is potential difference between the two.
(b) during this process energy will lose due to sparking.

2. At equilibrium F_{net} will be zero and at the extreme point, particle will be in state of rest.

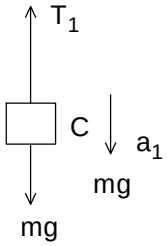
SECTION – C

1. in this situation block B will not move at all.

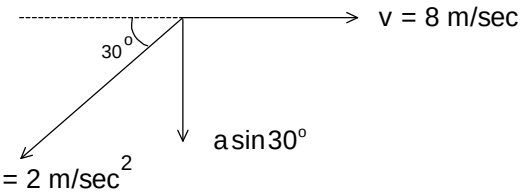


$T_2 = 2T_1$

Now apply newton's laws of motion.



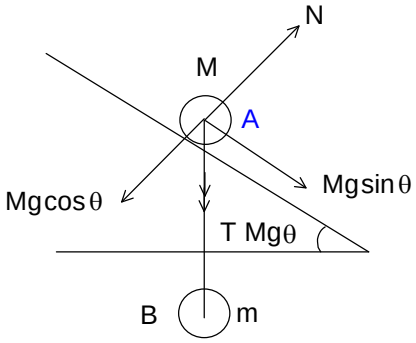
2. $a\sin 30^o = \frac{v_0^2}{R}$



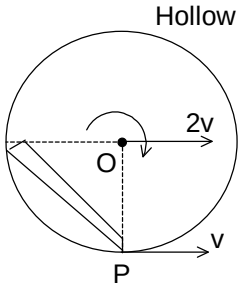
$a = 2$ m/sec^2

Where R is radius of curvature.

3. apply newton's laws of motion for the body A and body B



4. length of rod is $R\sqrt{2}$



Velocity of point P

$2v - \omega R = v$

KE of rod = rotational KE t translatory KE

5. simply use this concept
After earthing a conductor its potential will become zero.
If conductor are connected together both will be at the same potential.

6. from P – T diagram, write the P – T equation.
By using PV = nRT
Find P in terms of volume
$w = \int Pdv$.

Chemistry

PART – II

SECTION – A

1. C
Reimer – Tiemann's reaction

2. B
The cell reaction is: $H_2(g) + I_2(s) \rightleftharpoons 2H^+(aq) + 2I^-(aq)$

$$E_{cell} = E^o_{cell} - \frac{0.0591}{2}\log\frac{\left[H^+\right]^2\left[I^-\right]^2}{\left(P_{H_2}\right)}$$

$$0.7714 = 0.535 - \frac{0.0591}{2}\log\frac{\left[H^+\right]^2(0.1)^2}{1}$$

pH = 3

3. C
Mosley's equation is : $\sqrt{\upsilon} = a(z-b)$

$= az - ab$

Given that, ab = 1, a = $\tan 45^o = 1$

$\therefore \sqrt{\upsilon} = 51 - 1 = 50 \Rightarrow \upsilon = 2500$ Hz

4. A

$$S + 2Cl_2 \longrightarrow SCl_4 \xrightarrow{H_2O} \begin{matrix} HO & & OH \\ & S & \\ HO & & OH \end{matrix} \xrightarrow{-H_2O} HO - \overset{\cdot\cdot}{S} = O \\ \quad\quad\quad\

$Ag^{+}(aq) + Cl^{-}(aq) \longrightarrow AgCl(s)$

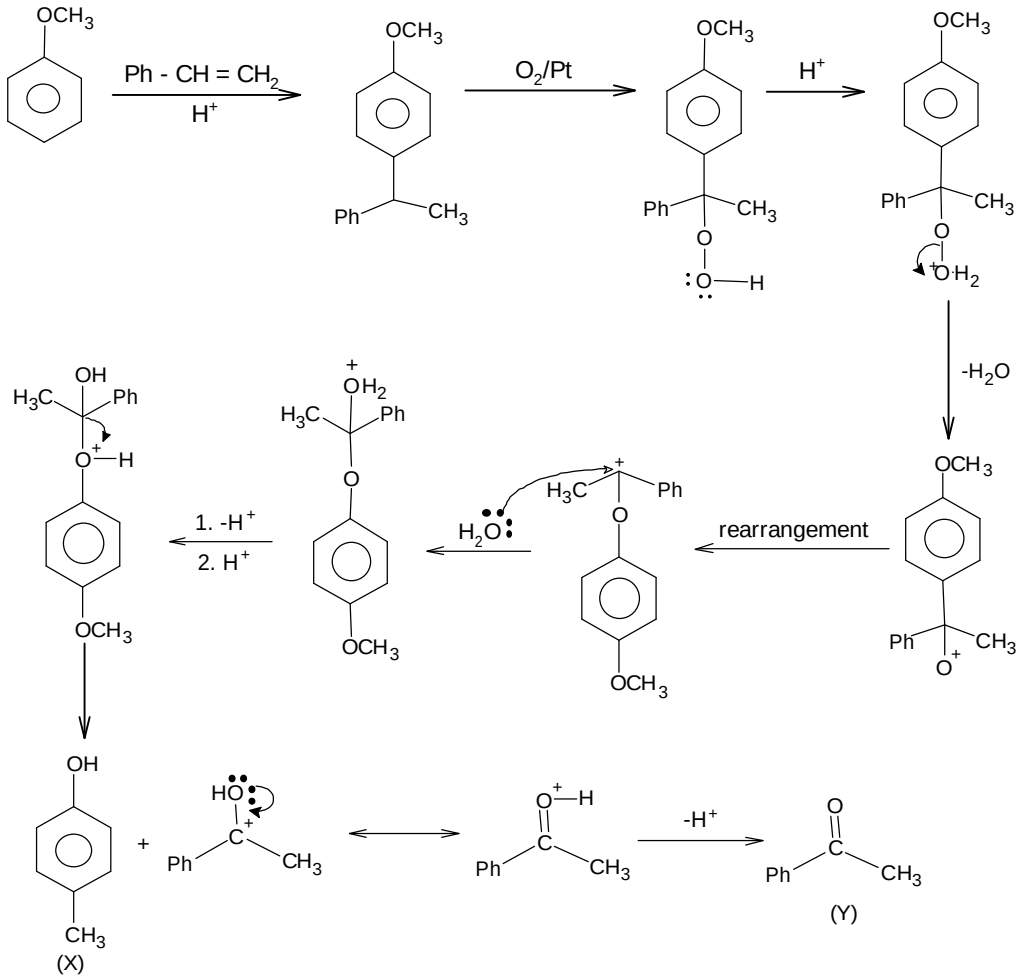
0.01 M	0.05	initial conc.
x	0.04 M	conc. after precipitation

$K_{sp} = \left[Ag^{+}\right]\left[Cl^{-}\right]$

$10^{-10} = \left[Ag^{+}\right](0.04)$

$\therefore \left[Ag^{+}\right] = 2.5 \times 10^{-9} M$

7. A



8. A, B

All reducing sugars are muatrotating. Although (IV) is an $\alpha -$ hydroxy ketone and hence reducing but it can't mutarotate as it not a carbohydrate, can't form ring. In (II) the glycosidic linkage is in between two anomeric carbons and hence ring opening can't occur, thus non – reducing as well as non – mutarotating.

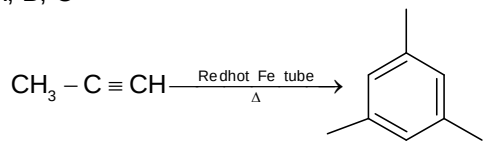
9. B, D

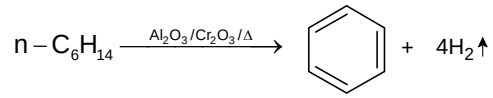
In B and D compound after mixing undergoes H – bonding and hence boiling point will be raised and hence vapour pressure lowered.

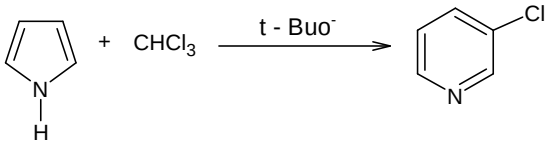
10. B, C

Fact

11. A, B, C







12. D

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + H_2O$$
(A) (B) (C)

$$N_2O + P_4 \longrightarrow P_4O_{10} + N_2$$
(D)

13. C

$$P_4O_{10} + HClO_4 \longrightarrow Cl_2O_7 + H_3PO_4$$

$$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$$

14. D

If k_1, k_2 and k_3 be the rate constants of the reaction along path I, II and III respectively, then overall rate constant of consumption of A will be $k_1 + k_2 + k_3$. So,

$$86.625 \text{ min} = \frac{0.693}{k_1 + k_2 + k_3}$$

$$\text{or } k_1 + k_2 + k_3 = \frac{0.693}{86.625} = 8 \times 10^{-3} \text{ min}^{-1}$$

$$k_1 = \frac{0.693}{173.25} = 4 \times 10^{-3} \text{ min}^{-1}$$

$$k_2 = \frac{0.693}{346.5} = 2 \times 10^{-3} \text{ min}^{-1}$$

$$\therefore k_3 = 8 \times 10^{-3} - (4 + 2) \times 10^{-3} = 2 \times 10^{-3} \text{ min}^{-1}$$

$$\%\text{distribution of C} = \frac{2k_2}{k_1 + 2k_2 + k_3} \times 100 = \frac{2 \times 2 \times 10^{-3}}{10 \times 10^{-3}} = 40$$

15. A

$$\frac{-d[A]}{dt} = (k_1 + k_2 + k_3)[A]$$

$$= 8 \times 10^{-3} \times 0.25 = 2 \times 10^{-3} \text{mol L}^{-1} \text{min}^{-1}$$

$$= \frac{d[B]}{dt} + \frac{d[C]}{dt} + \frac{d[D]}{dt}$$

$$= k_1[A] + 2k_2[A] + k_3[A]$$

$$= (k_1 + 2k_2 + k_3)[A] = 10 \times 10^{-3} \times 0.25 = 2.5 \times 10^{-3} \text{mol L}^{-1} \text{min}^{-1}.$$

16. C

$$E_{a(overall)} = \frac{k_1E_{a_1}}{k_1+k_2+k_3} + \frac{k_2E_{a_2}}{k_1+k_2+k_3} + \frac{k_3E_{a_3}}{k_1+k_2+k_3}$$
$$= 0.5\times 40 + 0.25\times 60 + 0.25\times 80$$
$$= 55\ kJ/mol$$

SECTION – B

1. $(A\rightarrow p, r, s)\ (B\rightarrow p, r, s, t)\ (C\rightarrow q, r)\ (D\rightarrow p, r, s)$

$3XeF_4 + 6H_2O \longrightarrow XeO_3 + 2Xe + \frac{3}{2}O_2 + 2HF$

$P_4 + 3OH^- + 3H_2O \longrightarrow PH_3 + 3H_2PO_2^-$

$B_2H_6 + 6H_2O \longrightarrow 2H_3BO_3 + 6H_2$

$B(OH)_3 + 2H_2O \longrightarrow \left[B(OH)_4\right]^- + H_3O^+$

$3Br_2 + 3Na_2CO_3 \longrightarrow 5NaBr + NaBrO_3 + 3CO_2$

2. $(A\rightarrow p, s)\ (B\rightarrow q, r, t)\ (C\rightarrow p, s)\ (D\rightarrow q, r, t)$
Fact

SECTION – C

1. 3

$_{92}U^{238} +_o n^1 \longrightarrow {}_{92}U^{239} \xrightarrow{-\beta} {}_{93}Np^{239} \xrightarrow{-\beta} {}_{94}Pu^{239}$

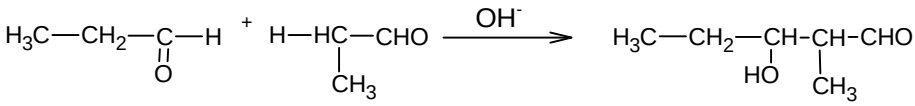
2. 2

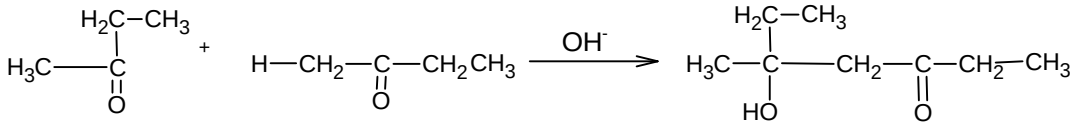
Effective no. of atoms of X = $4 - 4\times\frac{1}{8} = 3.5$

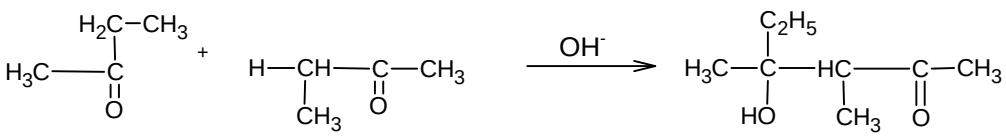
Effective no. of atoms of Y = 4 – 1 = 3
Effective no. of atoms Z = 8 – 4 = 4

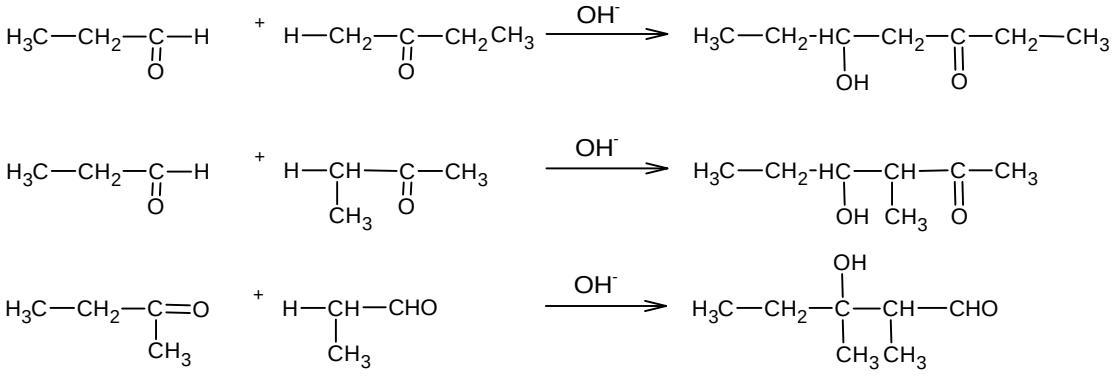
Therefore, required answer = $\frac{3+4}{\frac{7}{2}} = 2$

3. 6









4. 2

$\because \Delta U = nC_v\Delta T$

$12.5 = \frac{5.6}{22.4} \times C_v \times 10$

$C_v = \frac{50}{10} = 5$ cal

$\because C_p - C_v = R \Rightarrow C_p = 7$ cal

$\therefore \gamma = \frac{C_P}{C_V} = \frac{7}{5} = 1.4$

The gas is thus diatomic

5. 6

No. of eq. of NaOH that can be produced theoretically for 100% current efficiency = no of equivalents of NaCl decomposed = no of eq. of Cu deposited

$$= \frac{\text{wt. of Cu}}{\text{Eq. wt. of Cu}} = \frac{31.75}{63.5/2} = 1$$

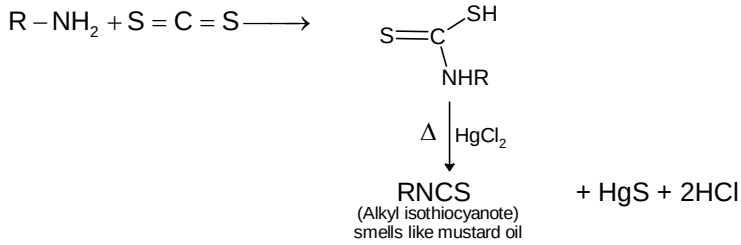
No. of eq. of NaOH produced experimentally = $\frac{\text{meq. of NaOH}}{1000}$

$$= \frac{N \times V}{1000} = \frac{1 \times 60}{1000} = 0.06$$

$\therefore$ % yield $= \frac{0.06}{1} \times 100 = 6\%$

6. 4

Hoffmann's mustard oil reaction is a test of 1° amine.



Mathematics

PART – III

SECTION – A

1. $f(x)=\dfrac{\int_0^n [x]dx}{\int_0^n \{x\}dx}$

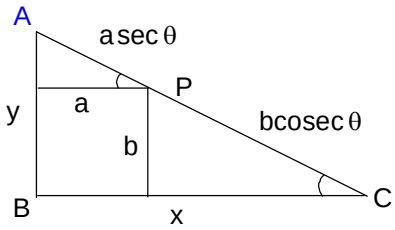
$\therefore \sum f(x)=\sum n-1=\dfrac{n(n+1)}{2}-n=\dfrac{n^2-n}{2}$

2. $\because y'y'''=3(y'')^2$

$\Rightarrow \int\dfrac{y'''}{y''}dx=3\int\dfrac{y''}{y'}dx$

$\therefore x=A_1y^2+A_2y+A_3$

3. $\dfrac{d(AC)}{d\theta}=-b\,\text{cosec}\,\theta.\cot\theta+a\sec\theta\tan\theta$



$AC=\dfrac{b\sqrt{a^{\frac{2}{3}}+b^{\frac{2}{3}}}}{b^{\frac{1}{3}}}+\dfrac{a\sqrt{a^{\frac{2}{3}}+b^{\frac{2}{3}}}}{a^{\frac{1}{3}}}$

$=\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{3}{2}}$

4. Let $\alpha,\beta,\gamma,\delta$ be the four roots

Then $\alpha+\beta+\gamma+\delta=4$ and $\alpha\beta\gamma\delta=1$

AM = GM

$\therefore \alpha=\beta=\gamma=\delta$

$\Rightarrow \alpha=\beta=\gamma=\delta=1$

$\therefore x^4-4x^3+ax^2+5x+1=(x-1)^4$

$\therefore a=6, b=-4$

5. Let APB = α

$\cos\alpha=\dfrac{PA^2+PB^2-AB^2}{2PA.PB}$

$\Rightarrow PA^2 + PB^2 - AB^2 = 2PA.PB\cos\alpha \le 2PA.PB\,(\text{as}\cos < 1)$

$\Rightarrow (PA - PB)^2 \le AB^2$

Thus the max value of |PA – PB| is AB

This is possible only when P lies on AB but P lies on AB

$\therefore$ P is the point of intersection of x + y = 2 and 2x + 3y + 1 = 0.

6. $\frac{6!}{P!q!r!}(2)^P(-x)^q(3x^2)^r$

$\because P + q + r = 6$ & $q + 2r = 5$

$\therefore$ coeff $= -5052$

7. $\tan 3\theta - \tan 2\theta = \tan 2\theta - \tan\theta$

$\Rightarrow \frac{\sin(3\theta - 2\theta)}{\cos 3\theta\cos 2\theta} = \frac{\sin(2\theta - \theta)}{\cos 2\theta\cos\theta}$

$\Rightarrow \sin\theta(2\sin\theta\sin 2\theta) = 0$

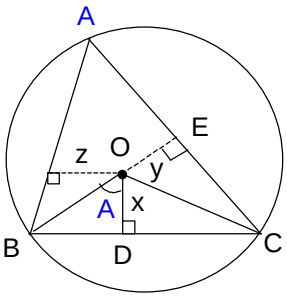
$\Rightarrow \sin\theta = 0$ or $\sin 2\theta = 0$

$\therefore \theta = n\pi, n \in z$ or $2\theta = n\pi$ $\because \theta = \frac{n\pi}{2}$ is not for possible as n is odd $\tan\theta$ is not define.

Hence $\theta = n\pi, n \in z$ is the only solution.

8. $\because AB = c, BC = a, AC = b$

In ΔOBD



$\cos A = \frac{OD}{OB}$

$\cos A = \frac{x}{R}$

$x = R\cos A$

$x = \frac{a}{2\sin A}\cos A \left(\because \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R\right)$

$x = \frac{a}{2}\cot A \ \therefore \tan A = \frac{a}{2x}$

Similarly

$\tan B = \frac{b}{2y}$

And $\tan C = \frac{c}{2z}$

Also $\tan A + \tan B + \tan C = \tan A.\tan B.\tan C$ (when A+ B + C = π)

$\Rightarrow \frac{a}{2x}+\frac{b}{2y}+\frac{c}{2z}=\frac{abc}{8xyz}$

$\Rightarrow \frac{a}{x}+\frac{b}{y}+\frac{c}{z}=\frac{abc}{4xyz}$

9. (A) $0<x<\frac{\pi}{4}$

$\tan x<\cot x$

also $\ln(\sin x)<0$

$(\tan x)^{\ln\sin x}>(\cot x)^{\ln(\sin x)}$

(B) in $\left(0,\frac{\pi}{2}\right)$

$\operatorname{cosec} x>1$

$4^{\ln\operatorname{cosec} x}<5^{\ln\operatorname{cosec} x}$

(C) in $\left(0,\frac{\pi}{2}\right)$, $\cos x\in(0,1)$

$\ln\cos x<0$

also $\frac{1}{2}>\frac{1}{3}$

$\left(\frac{1}{2}\right)^{\ln(\cos x)}<\left(\frac{1}{3}\right)^{\ln(\cos x)}$

10 $\because\ \ell = 5m - 3n$

$\therefore\ 7(5m-3n)^2+5m^2-3n^2=0$

$\Rightarrow 6m^2-7mn+2n^2=0$

$m=\frac{2}{3}n$ or $m=\frac{n}{2}$

If $m = 2/3n$ from (1) $\ell = 1/3\ n$

If $m = n/2$ ….. $\ell = -n/2$

$\therefore \cos\theta=\frac{1\times-1+2\times1+3\times2}{\sqrt{1^2+2^2+3^3}\sqrt{(-1)^2+1^2+2^2}}=\frac{7}{\sqrt{14}\sqrt{6}}=\frac{\sqrt{7}}{\sqrt{12}}$

11. $\left|\vec{a}+\vec{b}+\vec{c}+\vec{d}\right|^2=\sum\left|\vec{a}\right|^2+2\sum\vec{a}.\vec{b}=4+2\vec{d}.\left(\vec{a}+\vec{b}+\vec{c}\right)$ (a,b,c are mutually $\perp$ r)

Let $\vec{d}=\lambda\vec{a}+\mu\vec{b}+\nu\vec{c}$

Then $\vec{d}.\vec{a}=\vec{d}.\vec{b}=\vec{d}.\vec{c}=\cos\theta$

$\therefore \lambda=\mu=\nu=\cos\theta$

Also $\lambda^2+\mu^2+\nu^2=1=3\cos^2\theta$ or $\cos\theta=\frac{1}{\sqrt{3}}$

$\therefore \left|\vec{a}+\vec{b}+\vec{c}+\vec{d}\right|^2=4\pm2.\frac{3}{\sqrt{3}}$

$=4\pm2\sqrt{3}$

12. end point of latus rectum P(1, 2) & Q(1, -2)

Δ PAQ is isosceles right angled

$\therefore$ slope of PA is – 1
In equation $y - 2 = -(x - 1)$
$x + y - 3 = 0$
similarly equation of line QB
$x - y - 3 = 0$
solving $x + y - 3 = 0$ with parabola $y^2 = 4x$
$(3 - x)^2 = 4x$
$\therefore x = 1, 9$
$\therefore$ co-ordinate of B & C are (9, -6) & (9, 6) respectively
Area of trapezium
$= \frac{1}{2} \times (12 + 4) \times 8$
$= 64$ sq units

13. let the circumcenter of trapezium be T(n, 0)
Then PT = PB
$\sqrt{(n-1)^2 + 4} = \sqrt{(n-9)^2} + 36$
or n = 7
radius $\sqrt{40} = 2\sqrt{10}$

14. – 16 $z = -\alpha \pm \sqrt{\alpha^2 - 1}$
Case I when $-1 < \alpha < 1$
We get $\alpha^2 < 1$
$z = -\alpha \pm i\sqrt{1 - \alpha^2}$
or $x = -\alpha \Rightarrow y^2 = 1 - x^2$
or $x^2 + y^2 = 1$
case II $\alpha > 1$
$\alpha^2 - 1 > 0$
$z = -\alpha \pm \sqrt{\alpha^2 - 1}$
or $x = -\alpha \pm \sqrt{\alpha^2 - 1}, y = 0$
Roots are $\left(-\alpha + \sqrt{\alpha^2 - 1}, 0\right), \left(-\alpha - \sqrt{\alpha^2 - 1}, 0\right)$
One root lies inside the unit circle and other will lies outside the unit circle
Case III where α is very large then
$z = -\alpha - \sqrt{\alpha^2 - 1}$
$\cong -2\alpha$
$z = -\alpha + \sqrt{\alpha^2 - 1}$
$= \frac{1}{-\alpha - \sqrt{\alpha^2 - 1}} = -\frac{1}{2\alpha}$

SECTION – B

1. total number of function = $3^6 = 729$
(a) total number of onto function = ${}^3C_1 1^6 + {}^3C_2 2^6 + {}^3C_3 3^6$
$= 3 - 192 + 729$
$= 540$

(b) since $f(a_i) \neq b_i$
It means that a_1, a_2, a_3 cannot be assigned images b_1, b_2, b_3
Number of function = $2^3 3^3 = 216$
(c) number of invertible function = 0
as function is not one – one
(d) total many one function = 729 – 0
= 729

2. (a) $\sin^{-1}\sin(3\pi - 10)$

$= 3\pi - 10$

(b) $\cos^{-1}\cos(4\pi - 10)$

$= 4\pi - 10$

(c) $\tan^{-1} x + \tan^{-1}\dfrac{1}{x} = \dfrac{\pi}{2}$ if $x > 0$

$= -\dfrac{\pi}{2}$ if $x < 0$

(d) $\sin^{-1}\dfrac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n(n+1)}} = \tan^{-}\left(\dfrac{\sqrt{n} - \sqrt{n-1}}{1 + \sqrt{n(n-1)}}\right)$

$\therefore \sum_{n=1}^{\infty} \sin^{-1}\dfrac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n(n-1)}} = \sum_{n=1}^{\infty} \tan\left(\dfrac{\sqrt{n} - \sqrt{n-1}}{1 + \sqrt{n(n-1)}}\right)$

$= \tan^{-1}\sqrt{n}$

$= \dfrac{\pi}{2}$

SECTION – C

1. let each friend has n sons.
$\therefore$ 3 tickets can be distributed among 2n sons in ${}^{2n}C_3$ ways.
The number of ways distributing 3 tickets such that two tickets go to the sons of one and one tickets goes to sons of the other.
$= {}^{n}C_2 \times {}^{n}C_1 + {}^{n}C_1 \times {}^{n}C_2 = 2 \times {}^{n}C_1 \times {}^{n}C_2$
$\therefore$ probability that two tickets go to the sons of one and one tickets goes the sons of the other

$= \dfrac{2 \times {}^{n}C_1 \times {}^{n}C_2}{{}^{2n}C_3}$

$= \dfrac{6n^2(n-1)}{2n(2n-1)(2n-2)} = \dfrac{3n}{2(2n-1)}$

But from question

$\dfrac{6}{7} = \dfrac{3n}{2(2n-1)}$

$\therefore n = 4$
Hence total number of boys = 8.

2. the equation of the tangents to the circle $x^2 + y^2 = a^2$ at P and the hyperbola $x^2 - y^2 = a^2$ at Q are

$x + my = a\sqrt{1+m^2}$

or $x - my = a\sqrt{1-m^2}$ respectively

Where $y = mx$ is intersecting line through (2, 0)
Let (h, k) be the point of intersection of these two lines

$\therefore h+mk=a\sqrt{1+m^2}$ are $h-mk=a\sqrt{1-m^2}$

$\Rightarrow (h+mK)^2=a^2(1+m^2)$ and $(h-mK)^2=a^2(1-m^2)$

$\Rightarrow m^2(K^2-a^2)+2mhK+h^2-a^2=0$

$\Rightarrow m^2(K^2+a^2)-2mhK+h^2-a^2=0$

Elimination m from these two equation and we get

$\Rightarrow (a^4+4y^4)x^2=a^6$

3. $\because A+B+C=\pi$

Since $e^{i\pi}=-1$, $e^{i(B+C)}=e^{i(\pi-A)}=-e^{iA}\theta$

$$\because \Delta=\begin{vmatrix} e^{i2A} & -e^{i(A+B)} & -e^{i(A+C)} \\ -e^{i(A+B)} & e^{i2B} & -e^{i(B+C)} \\ -e^{i(A+C)} & -e^{i(B+C)} & e^{i2C} \end{vmatrix}$$

$$\Delta=\begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix}$$

$=-4$

$\therefore |\Delta|=|-4|=4$

4. put $\dfrac{x-1}{x+2}=t$

5. $OP<HP$

$\Rightarrow \sqrt{x^2+y^2}<3-x$

or $y^2<-6\left(x-\dfrac{3}{2}\right)$

Area of region = $\displaystyle\int_0^3 \frac{9-y^2}{6}dy$

= 3 sq unit

6. $f(x)=\lim\limits_{n\to\infty}\left(\dfrac{1\times 2\times 3\ldots\times n}{n\times n\times\ldots\times n}\right)^{\frac{1}{n}}$

$\log f(x)=\lim\limits_{n\to\infty}\dfrac{1}{n}\left(\log\dfrac{1}{n}+\log\dfrac{2}{n}+\ldots+\log\dfrac{n}{n}\right)$

$=\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{1}{n}\log\dfrac{r}{n}$

$=\displaystyle\int_0^1 \log x\,dx=-1$

$f(x)=e^{-1}=\dfrac{1}{e}$

Hence $e^2f(x)=e^2\times\dfrac{1}{e}=e$

$\left[e^2f(x)\right]=[e]=2$